Chapter 1. What is reinforcement learning?

This chapter covers

- A brief review of machine learning
- Introducing reinforcement learning as a subfield
- The basic framework of reinforcement learning

> *Computer languages of the future will be more concerned with goals and less with procedures specified by the programmer.*
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> *Marvin Minksy, 1970 ACM Turing Lecture*

If you're reading this book, you are probably familiar with how deep neural networks are used for things like image classification or prediction (and if not, just keep reading; we also have a crash course in deep learning in the appendix). *Deep reinforcement learning* (DRL) is a subfield of machine learning that utilizes deep learning models (i.e., neural networks) in reinforcement learning (RL) tasks (to be defined in section 1.2). In image classification we have a bunch of images that correspond to a set of discrete categories, such as images of different kinds of animals, and we want a machine learning model to interpret an image and classify the kind of animal in the image, as in figure 1.1.

Figure 1.1. An image classifier is a function or learning algorithm that takes in an image and returns a class label, classifying the image into one of a finite number of possible categories or classes.